

MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

Computer Science and Engineering Department

MINOR PROJECT SYNOPSIS

B.Tech VII Semester, Batch (2023-2027)

Team ID	MNP007
Project Title	Government Scheme Discovery & Eligibility Assistant: A Multi-Agent AI Framework for Citizen Empowerment
Track	Tech for Good / Public Infrastructure
Team Members	Keshav Jindal (Enrollment No. 00496402723) Vaibhav Garg (Enrollment No. 01196402723)
Project Guide	Dr. Yogesh Sharma

1. Title of the Project

Government Scheme Discovery & Eligibility Assistant

A Multi-Agent AI Framework for Citizen Empowerment that flips the paradigm from citizens searching for schemes to the system finding schemes for the citizen.

2. Problem Statement

Millions of citizens, especially those from marginalized or rural communities, fail to benefit from government welfare programs simply because they are unaware of their existence or confused by the bureaucratic language defining eligibility. There is a massive information asymmetry and a steep digital divide. Government scheme details are typically scattered across fragmented departmental websites, buried deep within dense PDF notifications, and written in complex legal or bureaucratic jargon. A citizen must manually cross-reference their personal demographics (income, caste, gender, occupation, location) against dozens of overlapping criteria just to figure out if they might qualify.

3. Why is the Particular Topic Chosen?

The core challenge in public welfare is not always a lack of resources, but a lack of accessible information. This topic was chosen because it sits at the critical intersection of "Tech for Good" and advanced AI. By leveraging Large Language Models (LLMs) and intelligent retrieval, a modest digital intervention can completely remove the burden of manual, bureaucratic research from the citizen, actively bridging the digital divide and ensuring welfare reaches those who need it most.

4. Objective(s) and Scope of the Project

Objectives:

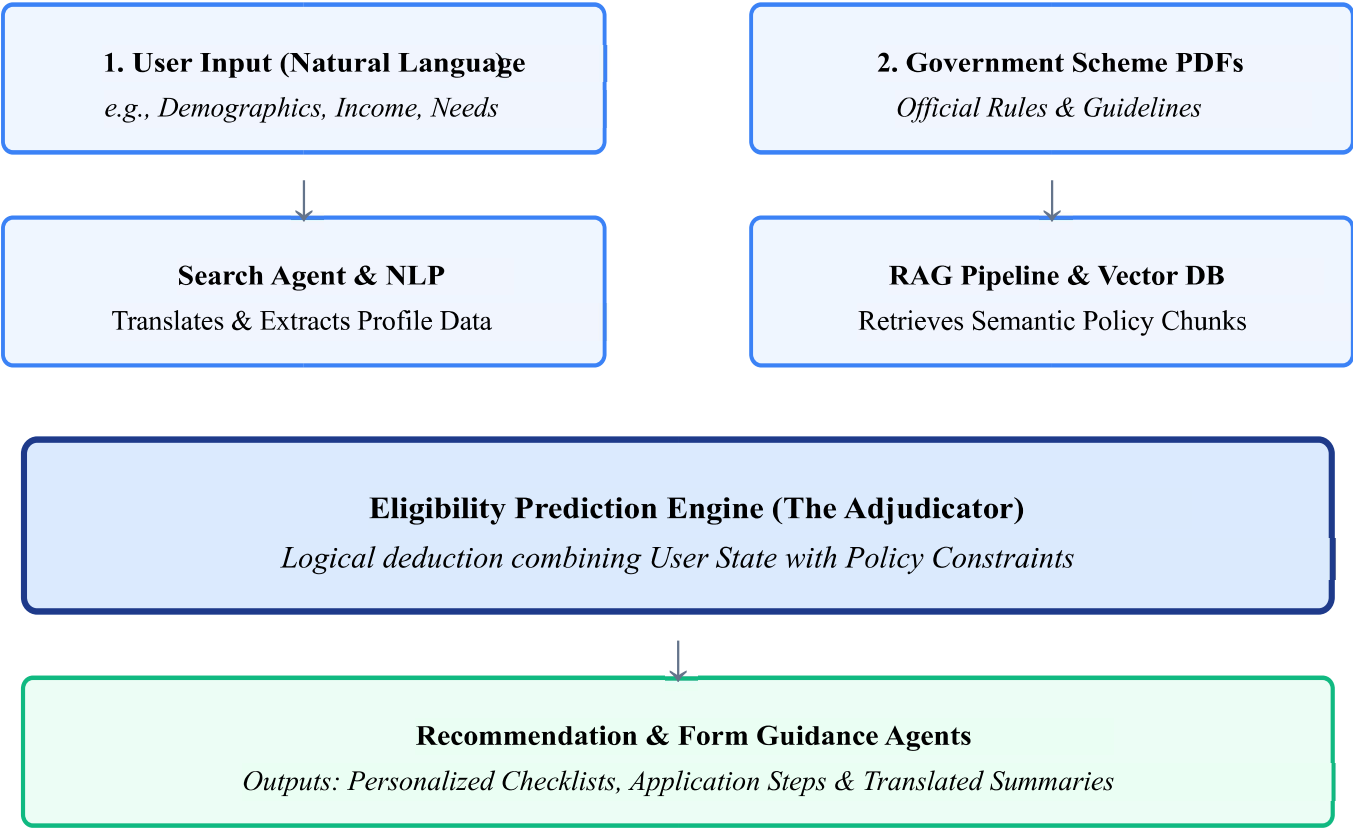
- **Understand Natural Language:** Allow citizens to explain their situation in their own words.
- **Process Complex Documentation:** Dynamically search through thousands of pages of official government PDFs to extract exact rules and benefits.
- **Determine Eligibility:** Programmatically match the extracted rules against the user's provided profile.
- **Explain & Guide:** Summarize the scheme in simple terms and provide a step-by-step roadmap for the application process, including required documentation.
- **Multilingual Support:** Support seamless language translation, allowing users to interact via text in regional languages.

Scope: The initial prototype focuses on conversational search, retrieval, and a multi-agent backend architecture (Search, Eligibility, Recommendation, and Form Guidance). Future scope and scalability include integrating directly with government APIs (like DigiLocker) to auto-verify documents, using Robotic Process Automation (RPA) to auto-fill forms, and deploying on platforms like WhatsApp for unprecedented scale.

5. Methodology, Process Description & Architecture

The architecture relies on four foundational AI pillars orchestrated by a Multi-Agent System to handle unstructured data and diverse user inputs:

- **RAG over Scheme PDFs:** Official government documents are parsed, chunked, and embedded into a vector database. Semantic search retrieves relevant paragraphs, ensuring the AI grounds its responses in official policy.
- **Search Agent (The Librarian):** Takes the user's query, formulates optimized searches, and retrieves the top-K relevant document chunks from the official scheme repository.
- **Eligibility Prediction Engine & Agent (The Adjudicator):** A hybrid reasoning engine that extracts explicit constraints (e.g., "Income < 1 Lakh") from the retrieved RAG context. It performs logical deduction to check off criteria against the user's state.



6. Hardware, Software, and Technology Stack

- **AI/ML Frameworks:** Large Language Models (LLMs) for reasoning and natural language processing.
- **Data Ingestion & Search:** Retrieval-Augmented Generation (RAG) architecture and Vector Databases for semantic search.
- **Processing Pipelines:** Advanced summarization pipelines to condense 20-page legal documents into 5-bulletpoint executive summaries.
- **Language & Orchestration:** Multilingual translation modules and a Multi-Agent orchestration framework for separating backend concerns.

7. Contribution Towards Society

This system fundamentally empowers marginalized citizens by bringing the government directly to their pocket. By removing language barriers through native-tongue translations and breaking down dense legal jargon into actionable, bite-sized summaries, it actively reduces poverty and inequality, ensuring that beneficial public infrastructure actually serves the public.

8. Schedule of the Project

Phase / Activity	Timeline
Requirement Analysis, UI Mockups & Data Sourcing	Weeks 1-2

RAG Setup: PDF Parsing, Chunking & Vector DB Implementation	Weeks 3-5
LLM Integration & Eligibility Prediction Engine Development	Weeks 6-8
Multi-Agent System Orchestration (Search, Recommend, Guide)	Weeks 9-11
Multilingual Chatbot Integration & Summarization Pipeline	Weeks 12-13
End-to-End System Integration, Testing & Bug Fixing	Weeks 14-15
Final Documentation, Poster Design & Defense Preparation	Week 16

9. Conclusion

The Government Scheme Discovery & Eligibility Assistant proposes a highly scalable, multi-agent AI solution to a critical public information failure. By combining RAG over official scheme PDFs with an eligibility prediction engine and multilingual chatbot capabilities, the system consolidates fragmented bureaucratic processes into a single, user-friendly conversational interface. This prototype successfully lays the groundwork for an automated, highly accessible digital public good that actively advocates for the citizen.

10. References / Bibliography

- [1] Lewis, P., et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems, 2020.
- [2] "LangChain Documentation: Multi-Agent Systems," [Online]. Available: <https://python.langchain.com>
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